WORKED SOLUTION

STEP III 1997, Question 7

Pure • Irrationality of e

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q7

The key idea

The series a_n is exactly what is left of $n!\,\mathrm{e}$ once you strip off its whole-number part: multiply the series $\mathrm{e} = \sum 1/k!$ by $n!$ and every term up to $k = n$ becomes an integer, while the tail becomes a_n. The geometric series b_n is there to act as an easy upper bound for a_n, term by term. Once we know the tail is strictly between 0 and 1, a rational e would make some $n!\,\mathrm{e}$ an integer with a non-zero fractional part, which is absurd.

Solution

Part (i)

The series b_n is geometric, with first term $\frac{1}{n+1}$ and common ratio $r = \frac{1}{n+1}$. Since $n \geq 1$ we have $0 < r \leq \frac{1}{2} < 1$, so it converges, and

$$b_n = \frac{\frac{1}{n+1}}{1 - \frac{1}{n+1}} = \frac{\frac{1}{n+1}}{\frac{n}{n+1}} = \frac{1}{n}.$$

Part (ii)

Write the kth terms of the two series as

$$u_k = \frac{1}{(n+1)(n+2)\cdots(n+k)}, \qquad v_k = \frac{1}{(n+1)^k}.$$

Each of the k factors $n+1, n+2, \ldots, n+k$ in the denominator of u_k is at least $n+1$, so $(n+1)(n+2)\cdots(n+k) \geq (n+1)^k$, and therefore

$$0 < u_k \leq v_k \quad \text{for every } k \geq 1.$$

Equality holds only for $k = 1$; for $k \geq 2$ the factor $n+2$ is strictly bigger than $n+1$, so $u_k < v_k$.

All the terms are positive and bounded by those of the convergent series b_n, so a_n converges (by comparison), and $a_n \geq u_1 > 0$. For the upper bound,

$$b_n - a_n = \sum_{k\geq 1}(v_k - u_k) \;\geq\; v_2 - u_2 = \frac{1}{(n+1)^2} - \frac{1}{(n+1)(n+2)} > 0,$$

because every term $v_k - u_k$ is non-negative. Hence $a_n < b_n = \frac{1}{n}$, and together

$$0 < a_n < \frac{1}{n},$$

as required.

Ask yourself Why isn't "each term of a_n is at most the corresponding term of b_n" enough to give a *strict* inequality?

Termwise $\leq$ only gives $a_n \leq b_n$ in the limit; strict inequalities can turn into equalities when you add infinitely many terms. That is why we pick out one term, $k = 2$, where the gap is a definite positive amount, and note that all the other gaps are ≥ 0.

Part (iii)

Using the series $\mathrm{e} = \sum_{k=0}^{\infty} \frac{1}{k!}$, multiply by $n!$ and split at $k = n$:

$$n!\,\mathrm{e} = \underbrace{\sum_{k=0}^{n} \frac{n!}{k!}}_{N} + \sum_{k=n+1}^{\infty} \frac{n!}{k!}.$$

For $k \leq n$, $\frac{n!}{k!} = (k+1)(k+2)\cdots n$ is an integer, so N is a positive integer. For $k = n + j$ with $j \geq 1$,

$$\frac{n!}{(n+j)!} = \frac{1}{(n+1)(n+2)\cdots(n+j)} = u_j,$$

so the tail is exactly a_n. Thus

$$n!\,\mathrm{e} = N + a_n, \qquad N \in \mathbb{Z}, \quad 0 < a_n < \tfrac{1}{n} \leq 1,$$

using part (ii) and $n \geq 1$. A number of the form (integer) + (something in $[0, 1)$) has that integer as its integer part, so $[n!\,\mathrm{e}] = N$, and therefore

$$a_n = n!\,\mathrm{e} - [n!\,\mathrm{e}].$$

Part (iv)

Suppose, for a contradiction, that $\mathrm{e} = \frac{p}{q}$ with p, q positive integers. Take $n = q$. Then

$$n!\,\mathrm{e} = q! \cdot \frac{p}{q} = (q-1)!\,p$$

is an integer, so it equals its own integer part, and part (iii) gives $a_q = 0$. But part (ii) says $a_q > 0$. This contradiction shows that e is irrational.

(Any $n \geq q$ would do equally well, since q divides $n!$; the point is only that some $n!\,\mathrm{e}$ would have to be a whole number.)

Checks

For $n = 1$: $1!\,\mathrm{e} = 2.71828\ldots$, so the fractional part is $0.71828\ldots$, and summing $\frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots$ gives $0.71828\ldots$ too, comfortably below $b_1 = 1$. For $n = 5$, $a_5 \approx 0.19382$, just under $\frac{1}{5}$: the bound gets relatively tighter as n grows, because the first terms of a_n and b_n agree. A computer check with exact fractions confirms $0 < a_n < \frac{1}{n}$ and $n!\,\mathrm{e} = N + a_n$ with $[n!\,\mathrm{e}] = N$ for every n from 1 to 29.

Common slips

- Claiming $a_n < b_n$ simply because $u_k \leq v_k$ for all k. You need one strictly smaller term (here $k = 2$) to keep the inequality strict.
- Forgetting why $\sum_{k \leq n} n!/k!$ is an integer: it is because $k!$ divides $n!$ when $k \leq n$.
- In part (iv), only using $0 < a_n$ or only $a_n < 1$. Both halves matter: $a_n < 1$ (from part (ii)) is what identifies a_n as the fractional part, and $a_n > 0$ is what gives the contradiction.

Beyond STEP

This is Fourier's proof that e is irrational, and its pattern (a quantity that would have to be an integer is shown to lie strictly between 0 and 1) is the template for many irrationality proofs, including those for π and e^r for rational $r \neq 0$. In a first analysis course you will justify the comparison test and the series for e properly, and see that the speed at which $\sum 1/k!$ converges is what makes the argument work: e is approximated by fractions far better than any rational number could be. Abbott's *Understanding Analysis* covers the series side, and Davenport's *The Higher Arithmetic* is a lovely introduction to rational approximation.

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